

Lab03

Socket programming practice with sample programs

BUPT/QMUL

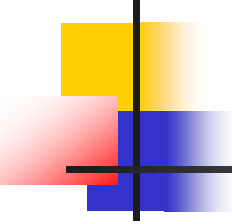
2026-4-9



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Tasks for Lab03

- **Task1:** Practice all the sample programs in Week2 and Week4 lectures
 - fork1.c
 - fork2.c
 - exec1.c
 - lseek1.c
 - readwrite1.c
 - udpechoct.c / udpechosvr.c

Lab04

Network Programming with UDP

BUPT/QMUL

2026-4-9



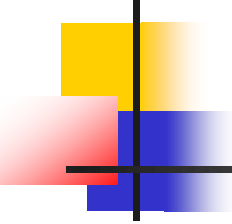
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要可以输入多个数据

server收到以后打印一些信息，比如数据来源和数据类型



Lab on UDP

- Write the programs for data sending based on UDP
 - You can design the running command format, e.g.,
 - ./<exefile> <server> <data1> <data2>
 - The server is specified in an IP address
 - **Multiple words** can be delivered each time
 - Each time the server receives data, **the source (*client's IP address and port number*) and the *data* should be displayed on the screen**

Note: You can complete this lab by modifying the code of *UdpEchoClnt.c* and *UdpEchoSvr.c* **to support multiple words transmission**. (Example is shown in the following)

An Example for your work

- Only an example, you can design the format by yourself

```
1 BUPTIA x 2 BUPTIA x +
student@BUPTIA:~/example$ ./UDPsrv
./UDPsrv: waiting for data on port UDP 1500
./UDPsrv: from 127.0.0.1:UDP35666 : hello,
./UDPsrv: from 127.0.0.1:UDP35666 : who
./UDPsrv: from 127.0.0.1:UDP35666 : are
./UDPsrv: from 127.0.0.1:UDP35666 : you?
```

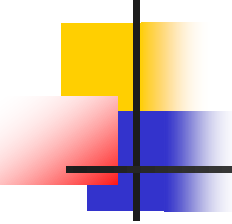
1. Run the server program in Terminal 1

3. Server receives the content and print it out.

```
1 BUPTIA x 2 BUPTIA x +
student@BUPTIA:~/example$ ./UDPclt 127.0.0.1 hello, who are you?
./UDPclt: sending data to '127.0.0.1' (IP : 127.0.0.1)
student@BUPTIA:~/example$
```

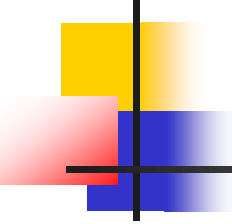
Input "IP" + "Content"

2. Run the client program in Terminal 2



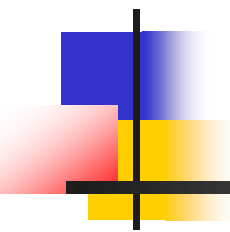
TIPs

- DO NOT forget the **#include <xxx.h>**
- Comments in **/* xxx */** can be ignored
- It is SEMICOLON “; ” but NOT COLON “: ” in the end of each line
- Open two Xshell terminals or use **Alt** + F1/F2 to switch terminals in Virtualbox.
- Run the **Server** firstly. Then run the **Client** and send data
- You can use “ifconfig” to check your IP address or use “**127.0.0.1**” as your IP address when you test the program
- **Try to read the compiling information when debug. Do not always beg help without any reading, thinking and analyzing.**



Lab04验收

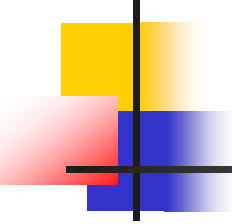
- 验收范围：全体，个人验收
- 验收截止时间：下次实验课（预计第9周）



选做Lab

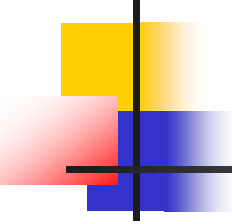
Finding DNS Information

BUPT/QMUL
2026-4-9



Tasks

- Write a program to find the DNS information about a given host
 - The host may be specified in domain name or IP address, e.g.,
 - ./<exefile> www.baidu.com
 - ./<exefile> 8.8.8.8
 - Use *gethostbyaddr()* and *gethostbyname()*
 - Your program shall list the official name, all the aliases, all the IP addresses in numbers-and-dots format
 - You can check your results by comparing with the output of "*nslookup*" command



一些提示

- 如何判断是根据domain name查IP还是根据IP查domain name?
 - Inet_aton()的返回值
- DNS查询后返回的是指向一个struct hostent的指针，如何引用struct中的分量？
 - Ptr->h_length
- 如何访问到alias列表和IP地址列表中的每个值？
 - for (p=ptr->h_alias; *p!=NULL; p++)
- 如何打印IP地址？
 - Inet_ntoa ()或inet_ntop ()
- 段错误

How to print IP address?

- Conversion between binary data (unsigned long) and dot-decimal notation

```
#include <arpa/inet.h>
```

Only support IPv4

```
char * inet_ntoa( struct in_addr in)
```

string(dot-decimal) **network address(binary)**

```
int inet_aton( const char * cp , struct in_addr * inp )
```

1- successful
0-failed